Exercise 1: Happy Supplies Parts Order

Original table

CustomerNumber	CustomerName	CustomerType	Date	Time	Employee	PartNumber	Name	Type	CageCode	QuantityOrdered	UnitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10654	Float Control	Plumbing	G413	4	4
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 AM	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

1(a). Assumptions

1. CustomerNumber determines CustomerName and CustomerType. PartNumber determines Name, Type, CageCode, and UnitPrice. An order is identified by CustomerNumber, Date, and Time because no order number is supplied. Each order has one employee, employees can serve any customer, and QuantityOrdered depends on the order and PartNumber.

1(b). Normalization process

1. UNF: The original form has a repeating group of ordered parts.

   PartsOrder(CustomerNumber, CustomerName, CustomerType, Date, Time, Employee, {PartNumber, Name, Type, CageCode, QuantityOrdered, UnitPrice})

2. 1NF: Put each ordered part on its own row so every field is atomic. PK: (CustomerNumber, Date, Time, PartNumber).

   PartsOrder_1NF(<u>CustomerNumber</u>, <u>Date</u>, <u>Time</u>, <u>PartNumber</u>, CustomerName, CustomerType, Employee, Name, Type, CageCode, QuantityOrdered, UnitPrice)

3. 2NF: Customer details depend only on CustomerNumber, and part details depend only on PartNumber. Separate these partial dependencies into relations:

   Customer_2NF(<u>CustomerNumber</u>, CustomerName, CustomerType)

   Part_2NF(<u>PartNumber</u>, Name, Type, CageCode, UnitPrice)

   OrderHeader_2NF(<u>CustomerNumber</u>, <u>Date</u>, <u>Time</u>, Employee)

   OrderLine_2NF(<u>CustomerNumber</u>, <u>Date</u>, <u>Time</u>, <u>PartNumber</u>, QuantityOrdered)

4. 3NF: No transitive dependencies remain under the stated assumptions. The final named relations are:

   Customer(<u>CustomerNumber</u>, CustomerName, CustomerType)

   Part(<u>PartNumber</u>, Name, Type, CageCode, UnitPrice)

   OrderHeader(<u>CustomerNumber</u>, <u>Date</u>, <u>Time</u>, Employee)

   OrderLine(<u>CustomerNumber</u>, <u>Date</u>, <u>Time</u>, <u>PartNumber</u>, QuantityOrdered)

PK/FK: OrderHeader.CustomerNumber references Customer. OrderLine (CustomerNumber, Date, Time) references OrderHeader; OrderLine.PartNumber references Part. Attributes that are both PK and FK are shown as both key roles in this description.

Exercise 2: Panacea Mental Health Corporation

Original table

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

2(a). Assumptions

1. staffNo determines therapistName; patNo determines patName. An appointment is identified by (patNo, appointmentDate, appointmentTime). Each appointment has one therapist and one branch. A therapist works at only one branch on a given date, so (staffNo, appointmentDate) determines branchNo.

2(b). Normalization process

1. UNF: The supplied relation has no repeating group shown; its attributes are listed below.

   TherapyAppointment(staffNo, therapistName, patNo, patName, appointmentDate, appointmentTime, branchNo)

2. 1NF: All values are atomic. PK: (patNo, appointmentDate, appointmentTime).

   TherapyAppointment_1NF(patNo, appointmentDate, appointmentTime, staffNo, therapistName, patName, branchNo); PK = (patNo, appointmentDate, appointmentTime)

3. 2NF: Patient and therapist names depend on their own identifiers, and branch assignment depends on therapist and date. Split the relations as follows:

   Patient_2NF(patNo, patName)

   Therapist_2NF(staffNo, therapistName)

   TherapistWorkday_2NF(staffNo, appointmentDate, branchNo)

   Appointment_2NF(patNo, appointmentDate, appointmentTime, staffNo)

4. 3NF: The final relations avoid repeating patient names, therapist names, and branch assignments:

   Patient(patNo, patName)

   Therapist(staffNo, therapistName)

   TherapistWorkday(staffNo, appointmentDate, branchNo)

   Appointment(patNo, appointmentDate, appointmentTime, staffNo)

PK/FK: TherapistWorkday.staffNo references Therapist. Appointment.patNo references Patient; Appointment.staffNo references Therapist; and (Appointment.staffNo, Appointment.appointmentDate) references TherapistWorkday. The latter FK uses staffNo and appointmentDate together.

Exercise 3: Maid Better Temp Agency

Original table

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

3(a). Assumptions

1. eNo determines eName. contractNo uniquely identifies a contract, and each contract applies to one event. eventNo determines eventLoc. Employees may work on multiple contracts and contracts may involve multiple employees; hours depends on (eNo, contractNo).

3(b). Normalization process

1. UNF: The supplied relation is EmployeeContract(eNo, contractNo, hours, eName, eventNo, eventLoc).

2. 1NF: All attributes are atomic. PK: (eNo, contractNo).

EmployeeContract_1NF(eNo, contractNo, hours, eName, eventNo, eventLoc)

3. 2NF: eName depends only on eNo, while eventNo and eventLoc depend on contractNo. Separate these partial dependencies:

Employee_2NF(eNo, eName)

Contract_2NF(contractNo, eventNo, eventLoc)

EmployeeContract_2NF(eNo, contractNo, hours)

4. 3NF: eventNo determines eventLoc, creating a transitive dependency in Contract_2NF. Separate event details:

Employee(eNo, eName)

Event(eventNo, eventLoc)

Contract(contractNo, eventNo)

EmployeeContract(eNo, contractNo, hours)

PK/FK: Contract.eventNo references Event. EmployeeContract.eNo references Employee and EmployeeContract.contractNo references Contract. Both EmployeeContract attributes are part of its composite PK and are also FKs.